

18.06 Problem Set 6

Due Friday, October 21 at 11am.

Problem 1 (5+5+5+5 points)

- (a) If P is the projection matrix onto the null space of A , then $Py - y$, for any y , is in the _____ space of A .
- (b) If $Ax = b$ has a solution x , then the closest vector to b in $N(A^T)$ is _____. (Try drawing a picture.)
- (c) If the rows of A (an $m \times n$ matrix) are independent, then the dimension $N(A^T A)$ is _____.
- (d) If a matrix U has orthonormal rows, then $I =$ _____ and the projection matrix onto the row space of U is _____. (Your answers should be simplified expressions involving U and U^T only.)

Problem 2 (5+10 points)

In class, we saw that the orthogonal projection p of a vector b onto $C(A)$ is given by $p = A\hat{x}$ where $\hat{x}$ is a solution to the "normal equations" $A^T A\hat{x} = A^T b$. We showed in class that these equations are *always* solvable, because $A^T b \in C(A^T) = C(A^T A)$.

- (a) The least-square solution $\hat{x}$ is unique if A is _____, in which case $A^T A$ is _____.
- (b) The least-square solution $\hat{x}$ is *not* unique if A is _____, in which case $A^T A$ is _____. However, the projection $p = A\hat{x}$ is *still* unique: if you have two solutions $\hat{x}$ and $\hat{x}'$ to the normal equations, $A\hat{x} - A\hat{x}' =$ _____ because _____.

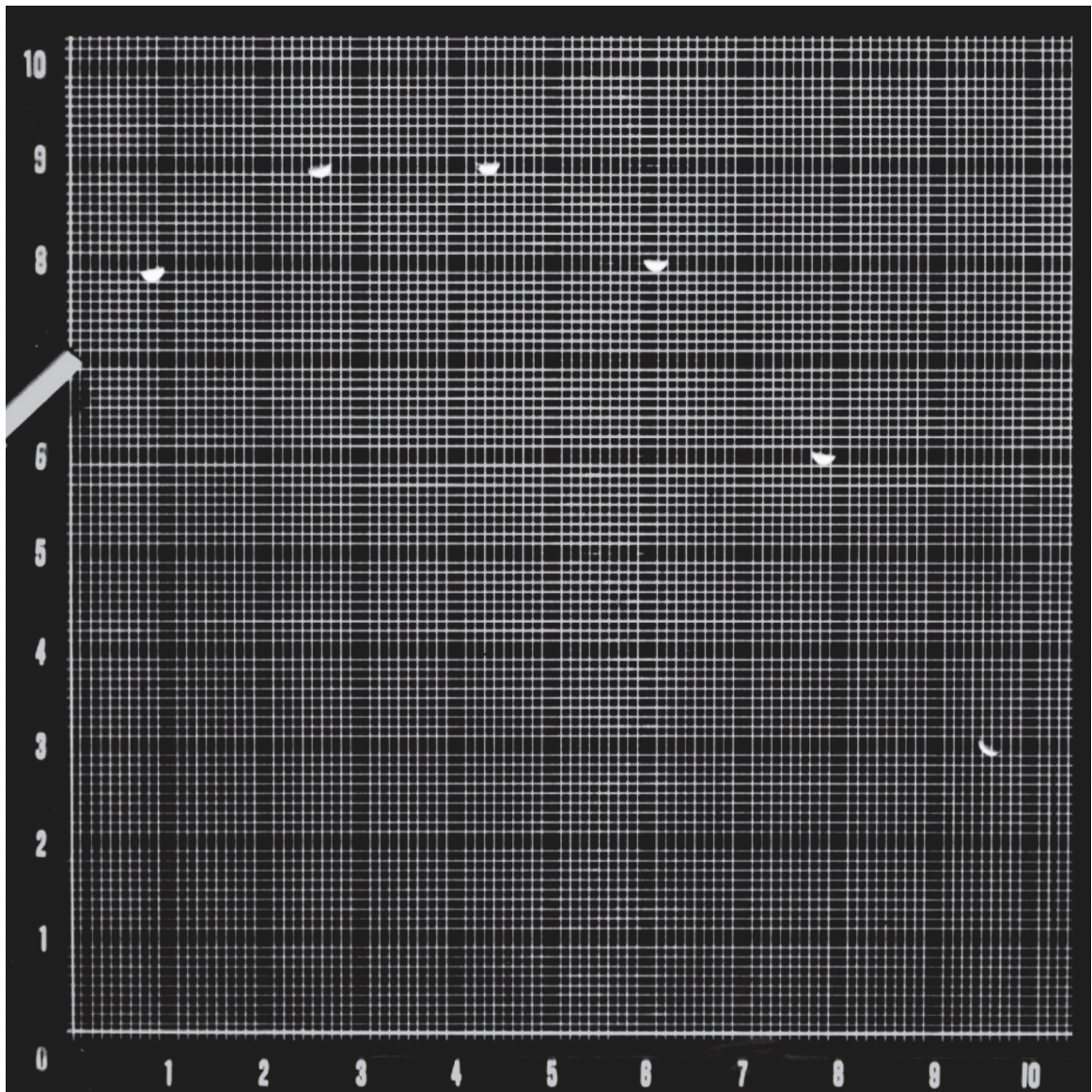
Problem 3 (5+5+10 points)

In this problem you will fit the motion of a projectile to a parabola in order to extract its velocity and the [Earth's gravitational acceleration](#) g .

If you launch a mass into the air from some position $x(0), y(0)$ with initial velocity $v_x(0), v_y(0)$, and friction is negligible, then you probably remember from 8.01 or high-school physics that the trajectory is a parabola:

$$x(t) = x(0) + v_x(0)t$$
$$y(t) = \underbrace{y(0) + v_y(0)t - \frac{g}{2}t^2}_{\text{parabola } y(t)} = y(0) + \underbrace{\frac{v_y(0)}{v_x(0)}[x - x(0)] - \frac{g}{2v_x(0)^2}[x - x(0)]^2}_{\text{parabola } y(x)}$$

The textbook [Physics2000](#) by E. R. Huggins contains [this strobe photograph](#) of a steel ball launched from the left, with a picture taken **every 0.1 seconds**. The spatial grid in the background is numbered in **10s of cm** (so that the numbers "1,2,3,...,10" refer to 10cm,20cm,30cm,100cm, and the fine grid lines are 1cm).



(a) Extract the data from the image for the 6 locations of the ball, and define Julia arrays `x = [...]`, `y = [...]`, and `t = [...]` holding the coordinates x, y, t (horizontal, vertical, time) of each ball location in **meters** and **seconds**, respectively. You can just zoom in and read the coordinates off the image by counting the fine grid lines, or use a fancier tool like [WebPlotDigitizer](https://webplotdigitizer.com/). Plot the data $x(t)$ and $y(t)$.

(b) Perform a least-square fit to $x(0) + v_x(0)t$ to extract the velocity $v_x(0)$. That is, define a 6×2 matrix `X` so that the least-square solution `X \ x` returns the 2-component vector $[x(0), v_x(0)]$ of the unknowns. Plot your linear fit along with the data $x(t)$.

(c) Perform a least-square fit to $y(0) + v_y(0)t - \frac{g}{2}t^2$ to extract the Earth's gravitational constant g in m/s^2 (you should get *roughly* 9.8m/s^2). That is, define a 6×3 matrix `Y` so that the least-square solution `Y \ y` returns the 3-component vector $[y(0), v_y(0), g]$ of the unknowns. Plot your parabolic fit along with the data $y(t)$.

A code outline is given below; fill in the blanks:

```
In [ ]: # part (a)

x = [...] # FILL IN
y = [...] # FILL IN
t = [...] # FILL IN

using PyPlot
plot(t, x, "bo-")
plot(t, y, "r*-")
xlabel("time (seconds)")
ylabel("(meters)")
legend([L"x(t)", L"y(t)"])
title("Problem 3a: Extracted x, y data")
```

```
In [ ]: # part (b)

X = [ ... ] # for example, [t.^3 t.^0 3t.^2] makes a matrix where each row
is  $t^3$  1  $3t^2$ 
xfit = X \ x # least square solution

@show x0 = xfit[1]
@show vx0 = xfit[2]

plot(t, x, "bo") # data

tf = range(t[begin], t[end], length=1000) # finely spaced times for
plotting curves
plot(tf, x0 .+ vx0 .* tf, "k-") # fit ... note that we use .+, .^, .* for
elementwise operations
legend(["data", "fit"])
xlabel("time (seconds)")
ylabel("(meters)")
title("Problem 3b: Linear fit of x(t)")
```

```
In [ ]: # part (c)

Y = [ ... ] # FILL IN, SIMILAR TO part (b)
yfit = Y \ y # least square solution

@show ... # SHOW YOUR FITTED y(0), vy(0), and g, similar to part (b)

plot(t, y, "r*") # data

... # PLOT THE PARABOLIC FIT and add labels, similar to part (b)
```

Problem 4 (10+5+5 points)

Suppose we have data (e.g. from some experimental measurement) $b_1, b_2, b_3, \dots, b_{21}$ at the 21 equally spaced times $t = -10, -9, \dots, 9, 10$. All measurements are $b_k = 0$ except that $b_{11} = 1$ at the middle time $t = 0$.

- (a) Using least squares, what are the best c and d to fit those 21 points by a straight line $c + dt$?
- (b) You are projecting the vector b onto what subspace? (Give a basis.)
- (c) Find a nonzero vector perpendicular to that subspace.

Problem 5 (10 points)

Suppose $\hat{x}$ is the least squares solution to $Ax \approx b$ (i.e. it minimizes $\|Ax - b\|$) and $\hat{y}$ is the least squares solution to $Ay \approx c$, where A has full column rank. Does this tell you the least squares solution $\hat{z}$ to $Az \approx b + c$? If so, what is $\hat{z}$ and why?

Problem 6 (5+5+5 points)

- (a) What matrix P projects every vector in $\mathbb{R}^3$ onto the line that passes through origin and $a = [3, 4, 5]$ (column vector)?
- (b) What is the nullspace of that matrix P ? (Give a basis.)
- (c) What is the row space of P^7 ?